

UIDAI Data Hackathon 2026

Unlocking Societal Trends in Aadhaar Enrolment and Updates

Analysis Report | Generated: January 11, 2026

1. Problem Statement & Approach

This analysis examines anonymized Aadhaar enrollment, demographic update, and biometric update datasets published by UIDAI to identify patterns, trends, anomalies, and predictive indicators. The objective is to uncover meaningful temporal, geographic, and demographic patterns, identify operational anomalies, and develop predictive insights that can support data-driven policy planning and capacity optimization.

UIDAI Data Hackathon 2026

Unlocking Societal Trends in Aadhaar Enrolment and Updates

Analysis Report | Generated: January 11, 2026

2. Datasets Used

2.1 Enrollment Dataset

Records analyzed: 319,549

Date range: 2025-01-04 00:00:00 to 2025-12-11 00:00:00

Geographic coverage: 40 states, 938 districts

Total enrollments: 2,622,028

2.2 Demographic Update Dataset

Records analyzed: 676,103

2.3 Biometric Update Dataset

Records analyzed: 871,726

UIDAI Data Hackathon 2026

Unlocking Societal Trends in Aadhaar Enrolment and Updates

Analysis Report | Generated: January 11, 2026

3. Methodology

- Data cleaning and validation
- Exploratory data analysis
- Statistical anomaly detection
- Machine learning based forecasting and anomaly detection
- All machine learning models were trained on the full aggregated dataset

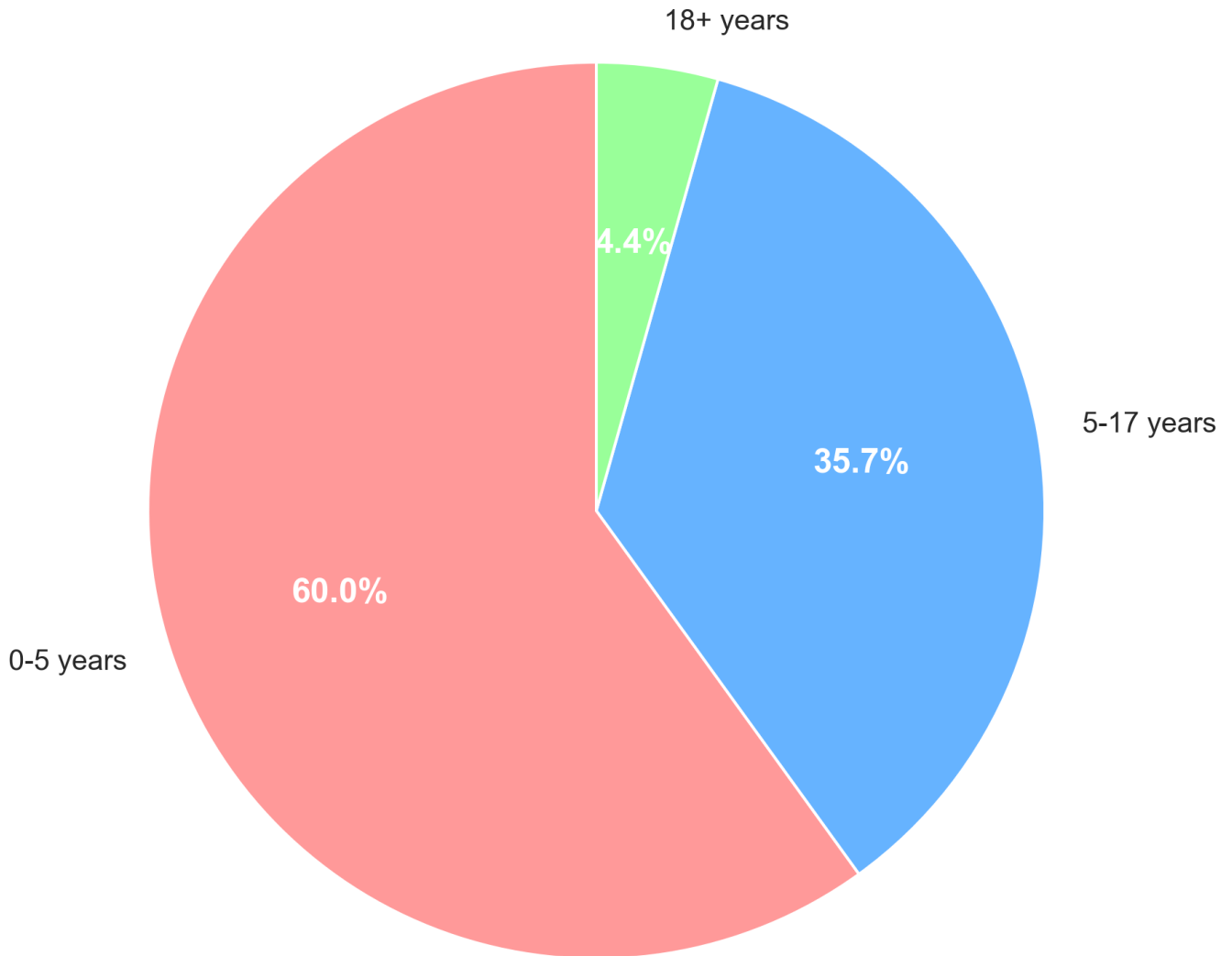
UIDAI Data Hackathon 2026

Unlocking Societal Trends in Aadhaar Enrolment and Updates

Analysis Report | Generated: January 11, 2026

4.1 Overall Age Distribution

Enrollment Distribution by Age Group

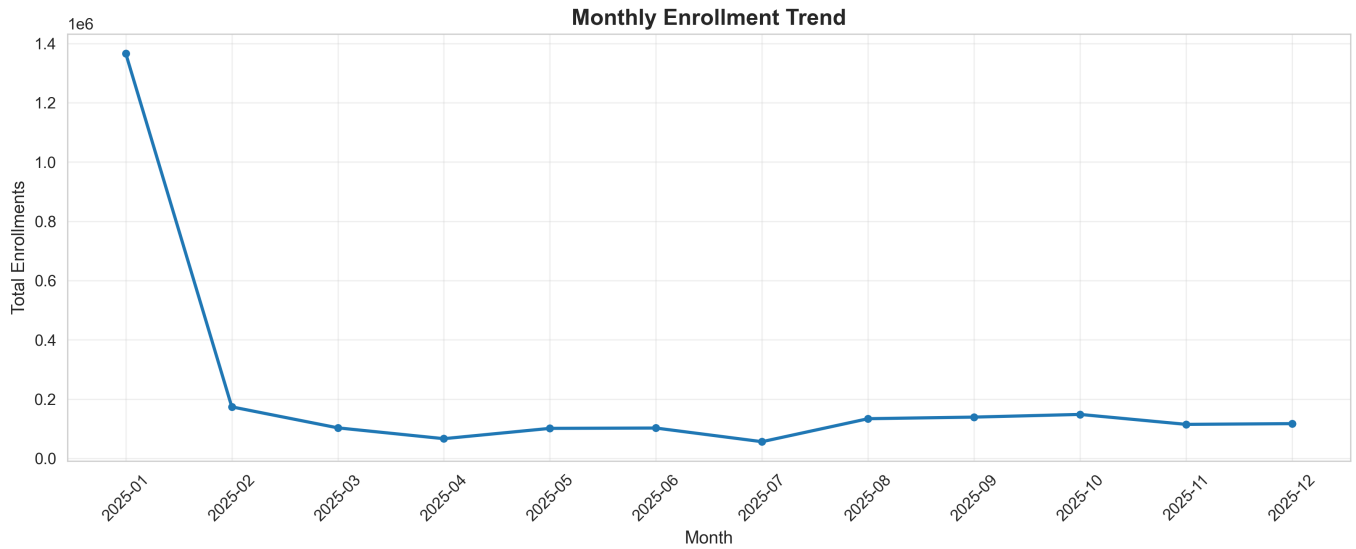


UIDAI Data Hackathon 2026

Unlocking Societal Trends in Aadhaar Enrolment and Updates

Analysis Report | Generated: January 11, 2026

4.3 Temporal Trends



UIDAI Data Hackathon 2026

Unlocking Societal Trends in Aadhaar Enrolment and Updates

Analysis Report | Generated: January 11, 2026

5. Key Findings & Insights

- Total enrollments analyzed: 2,622,028
- Geographic coverage: 40 states, 938 districts
- Age distribution: {'age_0_5': 59.96, 'age_5_17': 35.67, 'age_18_greater': 4.37}
- 2208 statistical anomalies detected
- Top performing state: Uttar Pradesh

UIDAI Data Hackathon 2026

Unlocking Societal Trends in Aadhaar Enrolment and Updates

Analysis Report | Generated: January 11, 2026

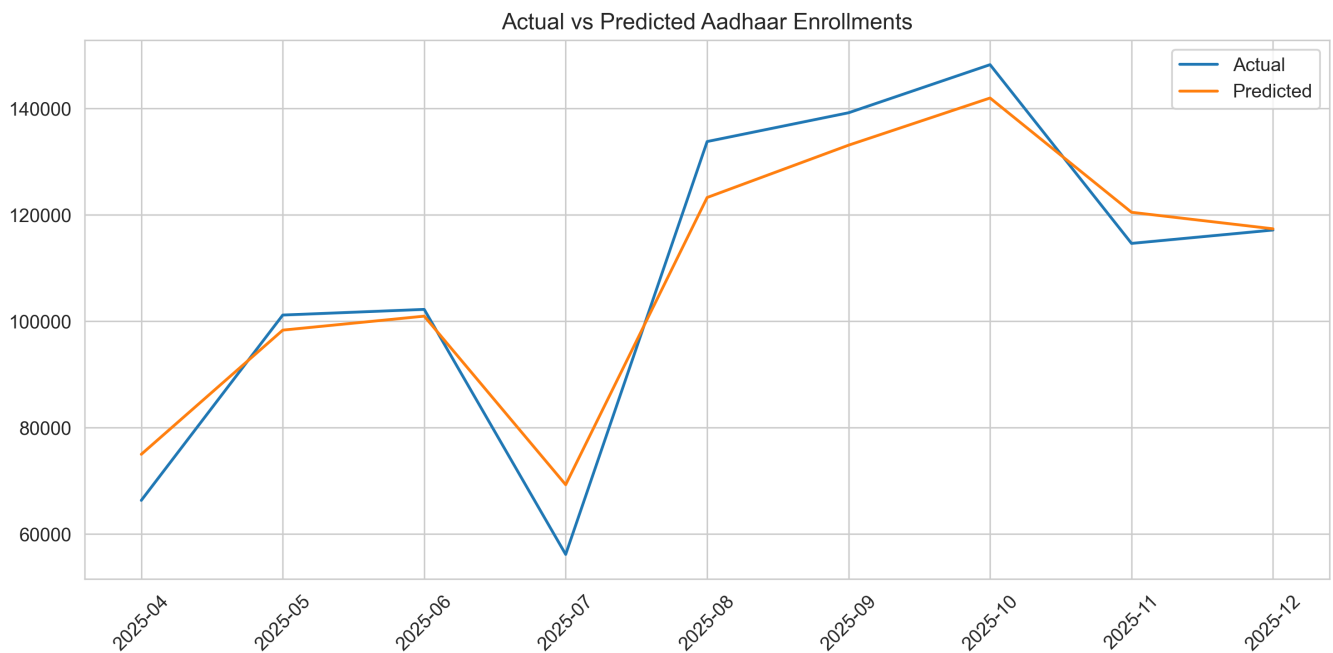
6. Machine Learning Analysis

6.1 Enrollment Forecasting

Random Forest model trained on full state-month data.

Mean Absolute Error (MAE): 190.09

States covered: 39



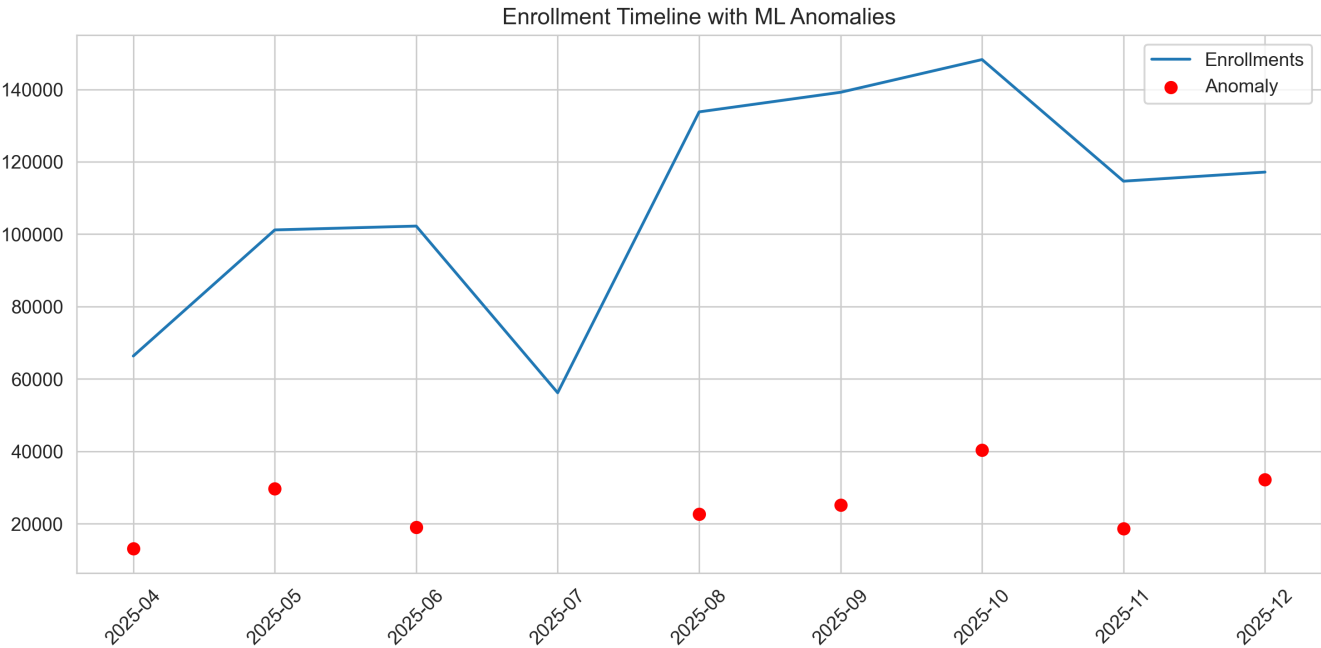
6.2 ML-based Anomaly Detection

Isolation Forest detected 11 anomalous state-month periods.

UIDAI Data Hackathon 2026

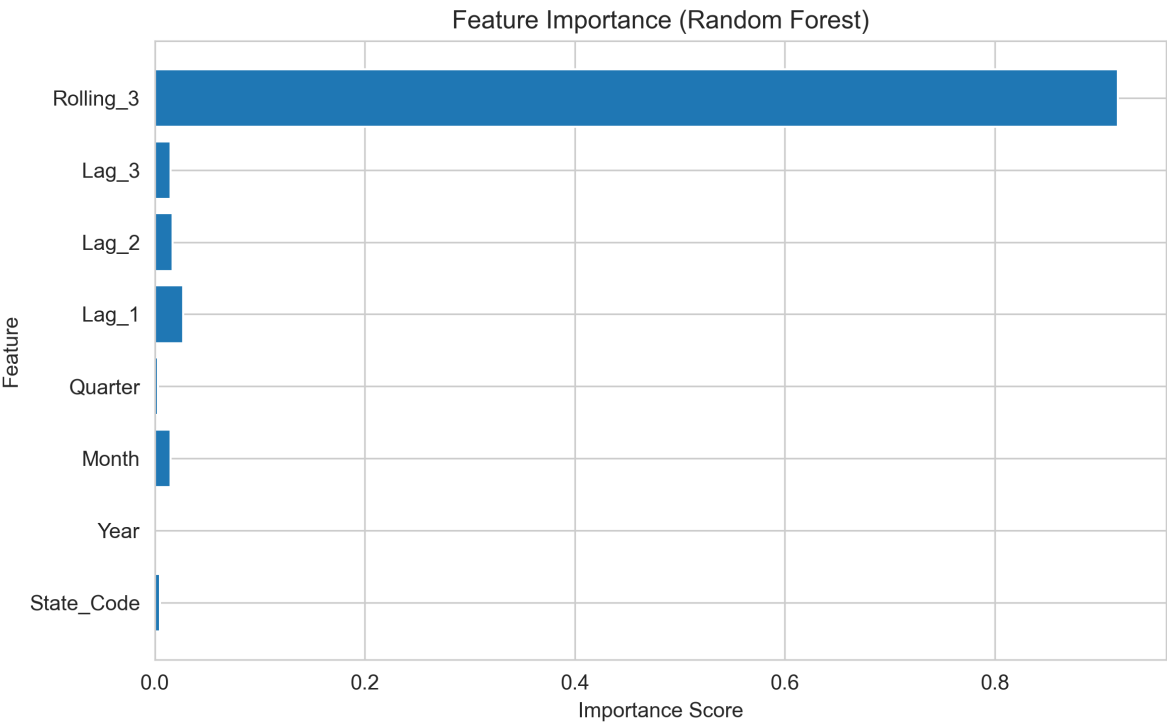
Unlocking Societal Trends in Aadhaar Enrolment and Updates

Analysis Report | Generated: January 11, 2026



6.3 Feature Importance

Feature importance analysis indicates that lag-based and rolling historical features are the strongest predictors, highlighting strong temporal dependence.



6.4 Policy Relevance

Machine learning enables predictive capacity planning and proactive monitoring of Aadhaar enrollment operations at a national scale.

UIDAI Data Hackathon 2026

Unlocking Societal Trends in Aadhaar Enrolment and Updates

Analysis Report | Generated: January 11, 2026

7. State Risk Profiling

States were classified into High, Medium, and Low risk categories based on enrollment volatility, growth instability, and machine learning detected anomalies. Only High and Medium risk states are shown to highlight regions requiring immediate attention or closer monitoring. Low risk states are excluded for clarity.

State	Risk	Avg Growth	Volatility	Anomalies
Uttar Pradesh	High	13.14%	5122	8
Madhya Pradesh	High	20.52%	3589	0
Bihar	High	22.49%	3125	3
Maharashtra	High	14.03%	2620	0
Rajasthan	High	18.27%	2580	0
West Bengal	High	11.85%	2266	0
Gujarat	High	28.33%	1884	0
Tamil Nadu	High	14.82%	1410	0
Karnataka	High	20.74%	1394	0
Jharkhand	High	34.34%	1388	0
Odisha	Medium	39.94%	1118	0
Telangana	Medium	16.80%	1082	0
Assam	Medium	16.08%	963	0
Andhra Pradesh	Medium	17.83%	769	0
Kerala	Medium	82.78%	752	0
Chhattisgarh	Medium	26.09%	723	0
Haryana	Medium	31.51%	678	0
Punjab	Medium	45.28%	575	0
Jammu And Kashmir	Medium	17.95%	390	0

UIDAI Data Hackathon 2026

Unlocking Societal Trends in Aadhaar Enrolment and Updates

Analysis Report | Generated: January 11, 2026

8. Conclusion

This study demonstrates how statistical analysis and machine learning together can support data-driven decision making for UIDAI.